

① What is Java and who developed it?

Ans:- Java is a object oriented programming language. It was developed by James Gosling at sun micro system, now owned by oracle.

② List any 5 features of java?

Ans:- Simple & Easy to learn:- it is easy to learn because its coding style is very clean & easy to understand.

Platform independence:- it is platform independent because, we write the code once but it runs on any platform.

OOP:- it supports object oriented programming because almost everything written in java is object & class, making it a true OOP.

Multithreading:- it supports multithreading because, when multiple programs runs automatically it's supported by Java.

Secure:- one of the most recognized feature is security because our java program runs inside JVM. it adds additional layer of protection.

③ Why is Java called platform independent?

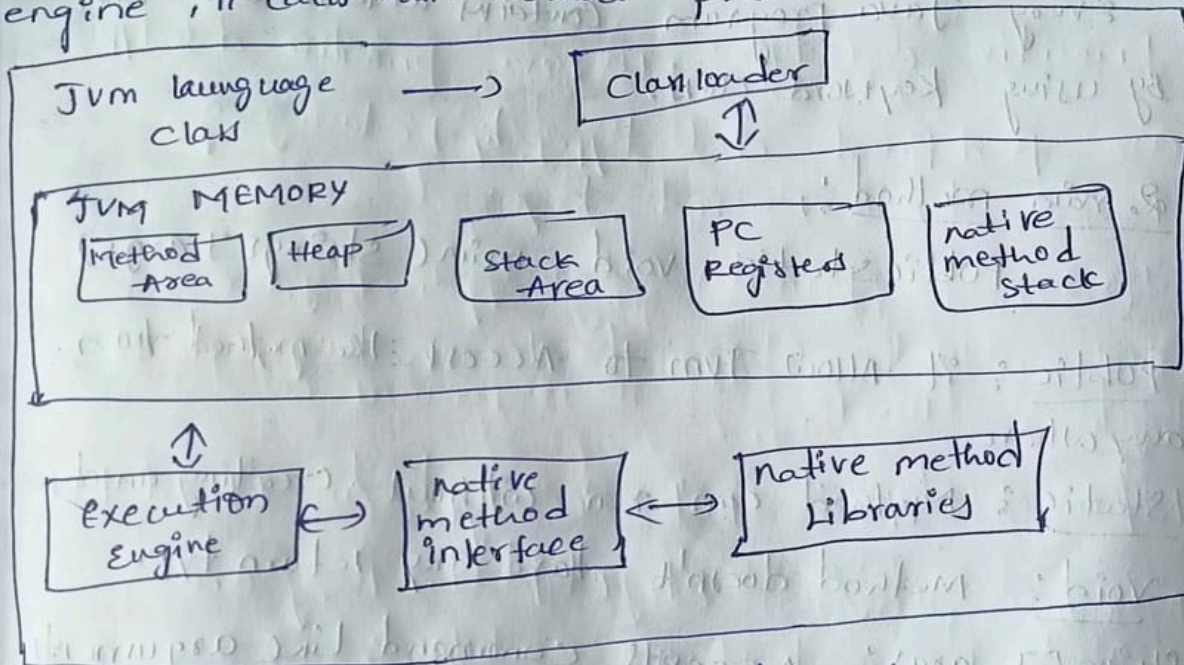
Java is platform independent because we write the code once, but its byte code will execute on any OS like macos, Linux, windows etc...

④ Difference between JDK, JRE, JVM

JDK	JRE	JVM
used to develop applications	used to run applications	executes Java byte code
it includes JRE + Development tools	JVM + libraries it contained.	it includes class loader, JIT compiler,
Writing & compiling Java code	Running a Java app ⁿ on a system	convert Byte code to machine code.

⑤ Architecture of JVM

JVM: it is used to runs Java app^s on runtime engine, it calls main method present in Java code.



⑥ What does role the JIT compiler play?

it helps to improve performance of java program by compiling bytecodes into machine readable code.

⑦ Basic Java program to print 'Hello, Java!'.

Ans:

```
public static  
public class Hello {  
    public static void main(String[] args) {  
        System.out.println("Hello, Java!");  
    }  
}
```

⑧ Explain each part of program.

1. class definition:

⇒ public class Hello {

// statements
}

Every Java program contains at least one class
by using keyword class.

2. main method:

```
public static void main(String[] args)
```

public: % Allow JVM to Access the method from
anywhere.

static: Method can run without creating an object.

void: Method doesn't have any return type.

String[] args: Accepts command line arguments.

3. System.out.println():

System :- Built in class from Java.lang package.

Out :- static member of System.

println() :- Method to print in console & moves to next line

⑨ What is the role of main() Method in Java?

Ans :- Java main() method is starting point from

Where JVM starts the execution of Java program.

→ JVM will not execute the program if main method is missing.

⑩ Define keywords in Java & with examples?

Sol :- Keywords are reserved words in Java, they have special meaning in Java, it tells to compiler what to do.

Ex :- "int" keyword tells to compiler declaring a variable of type integer, then it accepts only Integer type variable other than that it displays error.

⑪ List 10 keywords in Java.

int, return, boolean, do, catch, else,
float, char, case, try, finally, enum,
public, private, protected, super, this. While
throw, throws.

(12) What are identifiers & what rules to apply them.

→ identifier is the name given to variable, it is unique, may not be a keyword as identifier

rules to create identifier:

1. it allows all alphanumeric characters $([A-Z], [a-z], [0-9])$, '\$', and '-'.
2. identifier should not start with digits $[0-9]$.
3. identifier not be a keyword.

(13) What are primitive & reference datatypes in Java.

Primitive datatype: In Java primitive datatypes are predefined datatypes by the language named by a reserved keyword.

In Java 8 primitive datatypes they are:
int, float, char, boolean, long, double, byte, short.

Reference types: Reference types are stored the memory addresses of objects. They are

String

Array

Class

Interface

Object.

(14) Declare a variables of all primitive data types?

```
Ans:- byte myByte = 127;  
short myShort = 32767;  
int myInt = 2147483;  
long myLong = 922337203685L;  
float myFloat = 3.14f;  
double myDouble = 3.14159265d;  
char myChar = 'A';  
boolean myBoolean = true;
```

(15) Write a program to input & display student name, age, & grade?

```
Sol:- import java.util.*;  
  
public class Student {  
    public static void main (String[] args) {  
        Scanner sc = new Scanner (System.in);  
        System.out.println("Enter student age:");  
        int age = sc.nextInt();  
        System.out.println("Enter student name:");  
        String name = sc.next();  
        System.out.println("Enter grade:");  
        String grade = sc.next();  
    }  
}
```



```

        System.out.println("Name:" + name);
        System.out.println("Age:" + age);
        System.out.println("Grade:" + grade);
    }
}

```

⑩ Explain difference Between implicit & explicit casting?

implicit casting:- it converts ~~to~~ one datatype to another data type. it is automatically converted by machine.

Ex ② Example:-
 int num = 10;
 double numDouble = num;

③ there is no data loss during conversion.

④ used for converting compatible data types.

explicit type casting:- it is converts one data type to another, this is done by the end user. user forcibly changes the conversion.

② Example:-
 double pi = 3.14159;
 int pi = (int) pi;

③ There is may or maynot loss data during conversion.

④ used for converting incompatible data type.

(17) Write a program to demonstrate both type of Casting?

Ans:

```
public class TypeCasting {  
    public static void main (String[] args) {  
        int myInt = 20;  
        double myDouble = myInt;  
        float myFloat = 102.34f;  
        int myNew = (int) myFloat;  
        System.out.println ("Explicit Casting:" +  
                               myDouble);  
        System.out.println ("Explicit Casting:" + myNew);  
    }  
}
```

(18) What happens When narrowing Conversion fails?

Sol: narrowing conversion do not always succeeded at run time & can fail or incur data loss.

→ An error occurs when destination data type cannot receive the value being converted.

(19) Explain the difference between == and = operator.

Sol: "=" it assigns some value to variable
"==" it checks the addresses of 2 variables.

Q20) write a program using Arithmetic & Relational operators?

Sol: import java.util.*;

```
public class ArithmeticandRelational {  
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int a = sc.nextInt();
```

```
        int b = sc.nextInt();
```

```
        System.out.println("Addition: " + (a+b));
```

```
        System.out.println("Subtraction: " + (a-b));
```

```
        if (a > b) {
```

```
            System.out.println("a is greater");
```

```
        } else {
```

```
            System.out.println("b is greater");
```

```
        }
```

Q21) List & Explain logical operators in java.

→ logical and $\Rightarrow$ && $\Rightarrow$ used both condⁿs are true

logical or $\Rightarrow$ || $\Rightarrow$ either one condⁿ is true

Logical not $\Rightarrow$! $\Rightarrow$ if und true $\Rightarrow$ false,

false $\Rightarrow$ true.

(22) write a program to find the largest three numbers using if-else?

```
=> import java.util.*;

public class LargestThreeNumbers {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);

        int a = sc.nextInt();
        int b = sc.nextInt();
        int c = sc.nextInt();

        if (a > b && a > c) {
            System.out.println ("A is big" + a);
        }
        else if (b > c) {
            System.out.println ("b is big" + b);
        }
        else {
            System.out.println ("c is big" + c);
        }
    }
}
```

(23) demonstrate the use of java switch() Statement?

→ java switch statement is used to one statement from multiple conditions.


```

public class SwitchDemo {
    public static void main (String[] args) {
        int day = 2;
        switch (day) {
            case 1:
                System.out.println("Monday");
                break;
            case 2:
                System.out.println("Tuesday");
                break;
            default:
                System.out.println("Sunday");
        }
    }
}

```

(24) Write a loop that prints from 10 to 1.

Sol:-

```

public class TenToOne {
    public static void main (String[] args) {
        int i = 10;
        while (i > 0) {
            System.out.println(i);
            i--;
        }
    }
}

```


(25) What are arrays in Java

Arrays in Java are data structures that are used to store multiple values of same data types under single variable.

(26) Write a program to find sum of elements in an array?

sol:- public class sumArray {

public static void main (String[] args) {

int arr[] = {1, 2, 3, 4, 5};

int b = arr.length, int sum = 0;

for (int i = 0; i < b; i++) {

sum += arr[i];

System.out.println (sum);

(27) Declare and initialize a 2D array and print its elements..

public class TwoDArray {

public static void main (String[] args) {


```

int arr[] = { 1, 2, 3, 4, 5, 6, 7, 8, 9 };
for (int i = 0; i < arr.length; i++) {
    for (int j = 0; j < arr.length; j++) {
        System.out.print (arr[i] + arr[j] + " ");
    }
    System.out.println();
}
}

```